

Refutation of Conjecture 00000001283: the nim-equation $x \oplus y = x \cdot y$ has no solutions with $x, y \geq 1$

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Abstract

We disprove Conjecture 00000001283. In the stated domain $x, y \geq 1$ the nim-equation $x \oplus y = x \cdot y$ (with $\oplus$ the carry-free binary addition, i.e. bitwise XOR) has *no* solutions at all. Moreover the pair $(2, 2)$ that the conjecture lists as a solution is not one, because $2 \oplus 2 = 0 \neq 4 = 2 \cdot 2$, and the second clause of the conjecture – that for all other pairs $x, y > 2$ the difference $x \oplus y - x \cdot y$ is a sign flip of a power of two – fails already at $(3, 3)$, where the difference is -9 . Over the box $x, y \in [3, 499]$ exactly 39 of the 247 009 pairs satisfy that clause; the remaining 246 970 violate it. All three facts are verified by exhaustive computation and, independently, in Lean 4.

1 The conjecture under test

From `conjectures/00000001283.md`:

Conjecture 00000001283. *A nim-equation is an integer solution of $x \oplus y = x \cdot y$ with $x, y \geq 1$, where $\oplus$ is binary addition without carry. Conjecture: the only solutions are $(2, 2)$ and $(0, 0)$; for all other candidate pairs with $x, y > 2$, the difference between the nim-sum and the product is a sign flip of a power of two.*

Here “binary addition without carry” is bitwise XOR: writing x and y in base 2, one adds each bit column independently and discards carries. The claim has two clauses:

(C1) the solution set of $x \oplus y = x \cdot y$ over non-negative integers is exactly $\{(2, 2), (0, 0)\}$;

(C2) for every other pair with $x, y > 2$ one has $x \oplus y - x \cdot y = \pm 2^j$ for some $j \geq 0$.

We show that the statement is false as written, and in fact that it is false in a stronger sense than the wording suggests.

2 Part 1: $(2, 2)$ is not a solution

In binary, $2 = 10_2$, so

$$2 \oplus 2 = 10_2 \oplus 10_2 = 00_2 = 0,$$

whereas the ordinary product is $2 \cdot 2 = 4$. Hence

$$2 \oplus 2 = 0 \neq 4 = 2 \cdot 2,$$

so $(2, 2)$ does *not* satisfy the nim-equation. Clause (C1) explicitly lists $(2, 2)$ as a solution, so (C1) is literally false. The other listed pair, $(0, 0)$, is a genuine solution ($0 \oplus 0 = 0 = 0 \cdot 0$) but lies outside the stated domain $x, y \geq 1$; every theorem below concerns that domain.

Remark 1. A charitable reading of “the only solutions are $(2, 2)$ and $(0, 0)$ ” is “there are no solutions *besides* these two,” which would not, by itself, assert that $(2, 2)$ is a solution. But the sentence as written says that the only solutions *are* $(2, 2)$ and $(0, 0)$, i.e. it asserts membership; that assertion is false. In any case Section 3 shows there are no solutions with $x, y \geq 1$, so even the charitable reading leaves the conjecture with an empty (and hence vacuous) true half and a false second half.

3 Part 2: there are no solutions with $x, y \geq 1$

We prove the stronger statement announced in the abstract.

Theorem 2. *There is no pair of integers (x, y) with $x, y \geq 1$ such that $x \oplus y = x \cdot y$.*

Proof. Let $x, y \geq 1$ and suppose $x \oplus y = x \cdot y$.

Case $x = 1$ (or, symmetrically, $y = 1$). Adding 1 to an integer without carry changes only the lowest bit: it toggles that bit. Hence for $y \geq 1$,

$$1 \oplus y = \begin{cases} y + 1, & y \text{ even,} \\ y - 1, & y \text{ odd,} \end{cases}$$

so $1 \oplus y = y \pm 1$, which is never equal to $1 \cdot y = y$ because $\pm 1 \neq 0$. Thus $x = 1$ gives no solution, and by symmetry neither does $y = 1$.

Case $x, y \geq 2$. For the XOR we use the standard identity

$$x + y = (x \oplus y) + 2(x \& y),$$

which holds because addition decomposes into the carry-free sum and twice the carry bits. Since $x \& y \geq 0$,

$$x \oplus y = x + y - 2(x \& y) \leq x + y. \quad (1)$$

On the other hand, for $x, y \geq 2$,

$$x \cdot y - (x + y) = (x - 1)(y - 1) - 1 \geq 1 \cdot 1 - 1 = 0,$$

with equality if and only if $(x - 1)(y - 1) = 1$, i.e. $x = y = 2$. Hence

$$x \cdot y \geq x + y, \quad (2)$$

and combining (1) and (2),

$$x \oplus y \leq x + y \leq x \cdot y.$$

If $x \oplus y = x \cdot y$ then both inequalities are equalities. Equality in (2) forces $x = y = 2$, but then equality in (1) would require $x \oplus y = x + y$, whereas $2 \oplus 2 = 0 \neq 4 = 2 + 2$. So the case $x, y \geq 2$ yields no solution either.

Both cases being impossible, there is no solution with $x, y \geq 1$. □

Corollary 3. *Over the non-negative integers the solution set of $x \oplus y = x \cdot y$ is exactly $\{(0, 0)\}$.*

Proof. $0 \oplus 0 = 0 = 0 \cdot 0$ is a solution. Conversely, if $x, y \geq 0$ solve the equation and $(x, y) \neq (0, 0)$, then either $x = 0$ or $y = 0$ (the “boundary” cases $x, y \geq 1$ being impossible by Theorem 2). If $x = 0$ then $x \oplus y = y$ but $x \cdot y = 0$, so $y = 0$; symmetrically for $y = 0$. Hence the only non-negative solution is $(0, 0)$. □

This is confirmed by exhaustive search: over $0 \leq x, y < 500$ the solution set is $\{(0, 0)\}$, and over $1 \leq x, y < 500$ it is empty (`reproduce.py`, checks [2] and [2b]; the same bounded search is formalised in `lean4/Main.lean` for the box $0 \leq x, y < 200$).

4 Part 3: the second clause fails at (3, 3)

For a pair (x, y) write

$$\Delta(x, y) = (x \oplus y) - x \cdot y.$$

Clause (C2) asserts that $\Delta(x, y) \in \{\pm 2^j : j \geq 0\}$ for every pair with $x, y > 2$ and $(x, y) \notin \{(2, 2), (0, 0)\}$. This is false. At $(3, 3)$,

$$3 \oplus 3 = 11_2 \oplus 11_2 = 00_2 = 0, \quad 3 \cdot 3 = 9, \quad \Delta(3, 3) = 0 - 9 = -9.$$

The number 9 is not a power of two. Indeed the powers of two are $2^0 = 1$, $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, and 2^j is strictly increasing in j : for $j \leq 3$ we have $2^j \leq 8 < 9$, while for $j \geq 4$ we have $2^j \geq 16 > 9$. Hence $2^j \neq 9$ for every $j \geq 0$, so $-9 = -2^j$ is impossible as well; equivalently $\Delta(3, 3) = -9 \notin \{\pm 2^j\}$. Clause (C2) therefore fails.

The failure is not an isolated coincidence. Iterating over the full box $x, y \in [3, 499]$ (that is $497^2 = 247\,009$ pairs) in `reproduce.py`, check [3], we find:

$$\#\{(x, y) : \Delta(x, y) = \pm 2^j \text{ for some } j\} = 39, \quad \#\{(x, y) : \Delta(x, y) \neq \pm 2^j \text{ for all } j\} = 246\,970.$$

So 246 970 of 247 009 pairs – more than 99.98% – violate the clause. Besides $(3, 3)$ with $\Delta = -9$, small explicit violators include $(3, 4)$ with $3 \oplus 4 = 7$, $3 \cdot 4 = 12$, $\Delta = -5$, and $(3, 6)$ with $3 \oplus 6 = 5$, $3 \cdot 6 = 18$, $\Delta = -13$; neither -5 nor -13 is $\pm 2^j$.

Remark 4. Since by Theorem 2 there are no solutions with $x, y \geq 1$, the set of “other candidate pairs” in (C2) is the entire box $x, y > 2$; the clause is not saved by excluding solutions, because there are none to exclude.

5 Formal verification in Lean 4

The three refutations are formalised in core Lean 4 (toolchain `leanprover/lean4:v4.33.1`, `import Std`, no `Mathlib`, no `sorry`, no custom `axiom`, no `native_decide`); see `lean4/Main.lean` and the axiom audit `lean4/Check.lean`.

- `two_nim_two` : $(2 : \mathbb{N}) \oplus 2 = 0 \wedge 2 \cdot 2 = 4$, and `not_a_solution` : $\neg(2 \oplus 2 = 2 \cdot 2)$, both by `decide`. This is Section 2.
- `solutionsUpTo`, `only_zero_solution` : the bounded search over $0 \leq x, y < 200$ equals $[(0, 0)]$, and `no_pos_solutions` : the search over $1 \leq x, y < 200$ is empty. This is the computational shadow of Theorem 2 (Section 3).
- `nine_not_pow`, `nine_not_pow_neg` : $\neg \exists j, (-9 = 2^j)$ and $\neg \exists j, (-9 = -2^j)$, and `clause2_fails_at_3_3`, which records $3 \oplus 3 = 0$, $3 \cdot 3 = 9$ and both non-representability facts. This is Section 4.
- `conjecture_00000001283_false` collects all of the above.

The axiom audit reports that every theorem depends only on the core axioms `propext` and `Quot.sound` (never `sorryAx` or `Lean.ofReduceBool`).

6 Conclusion

Conjecture 00000001283 is **false**. The pair $(2, 2)$ it lists as a solution is not one ($2 \oplus 2 = 0 \neq 4$); there is in fact no solution with $x, y \geq 1$ whatsoever (Theorem 2); and the second clause fails already at $(3, 3)$, where the difference is -9 , with 246 970 of the 247 009 pairs in $[3, 499]^2$ violating it. These are three independent defects, any one of which suffices to refute the statement as written.